Requirements Table

Internal Microservice Catalog & Health Portal

Lab 1: Requirements Engineering & UML Use-Case Modelling

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall ping microservice health endpoints at 30-second intervals and compute rolling 24-hour service availability percentages.	**High**	Pass: Downtime is recorded and an alert is dispatched on 3 consecutive failed health probes. Fail: An unreachable service is marked healthy.	Core monitoring capability that enables early detection of outages.
FR-002	Functional	The system shall render an interactive dependency graph showing the upstream and downstream relationships for every registered microservice.	**High**	Pass: Selecting any service node highlights all direct dependencies and dependents within 1 second. Fail: The graph fails to load or a relationship is missing/incorrect.	Lets engineers quickly assess the blast radius of a service outage.
FR-003	Functional	The system shall allow a DevOps Engineer to search aggregated API documentation by service name, endpoint path, or HTTP method.	**Medium**	Pass: A search for a known endpoint returns the correct service's documentation within 3 seconds. Fail: The search returns no result or the wrong service's docs.	Reduces time spent hunting for API contracts across teams.
FR-004	Functional	The system shall allow a System Architect to register a new microservice with its name, owning team, health-check endpoint URL, and source-repository link.	**High**	Pass: After submission, the new service appears in the catalog and dependency graph within 1 minute. Fail: The service is not listed, or required fields are not validated.	Keeps the catalog accurate and complete as new services are deployed.
FR-005	Functional	The system shall dispatch a downtime alert to the on-call DevOps Engineer's configured channel when a monitored service records 3 consecutive failed health probes.	**Medium**	Pass: An alert is received in the configured channel within 60 seconds of the 3rd consecutive failure. Fail: No alert is sent, or one is sent before the 3rd failure.	Ensures timely incident response and minimizes downtime.
NFR-001	Nonfunctional (Performance & Security)	The service catalog's dependency-graph viewer must render interactive node architectures containing up to 200 services smoothly.	**High**	Pass: Benchmark tests confirm target render latency (< 2s) and security standards are met under simulated peak load. Fail: Rendering exceeds latency target or a security check fails.	Ensures usability at enterprise scale without degrading engineer productivity.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-002	Nonfunctional (Reliability & Availability)	The health-monitoring portal itself shall maintain at least 99.9% uptime measured monthly, and shall retain all health-probe logs for at least 90 days.	**High**	Pass: Monthly uptime reports confirm ≥ 99.9% availability and probe logs are retrievable for any date within the last 90 days. Fail: Uptime falls below threshold or logs are purged early.	The monitoring system must be more reliable than the services it observes; logs support incident post-mortems and audits.